

Applications of machine learning:

- Used for Email spam Detection
- Predictions like these price prediction
- Credit Fraud detections.
- Disease and symptoms prediction, etc.

Applications of Deep learning:

- Used in Face recognition system
- Automatic or Self-driving cars
- Also used in computer vision A

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Class - Test

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① Features

Machine Learning

Deep Learning

Architecture

uses simple algorithms
like Decision tree, SVM
Random Forest

uses Artificial
Neural Networks
with multiple
hidden layers

Data requirement

Performs well with
small to medium datasets

Requires very
large datasets

Feature Extraction

Manual feature Engineering manually
is required
learning
features

Training Time

Faster

Slower due to
complex computations

can Handle more

• Can run on CPU

usually requires
GPU/TPU

Accuracy

Good for structured
data

excellent for
complex data
like images
audio and text

Influence of Depth :-

Depth = Number of hidden layers.

Advantages :-

- learns hierarchical features.
- Higher accuracy on Images, speech and NLP.

Disadvantages :-

- Takes longer time to be trained.

Explain the concept of Representation Learning. Discuss how the width and depth of a NN influence its learning capability, computation complexity and model performance.

Q Representation Learning:

Representation Learning is a technique where a neural network automatically learns useful features from raw input data without manual feature engineering.

Example: Instead of manually detecting edges in a image, a CNN learns this automatically.

Influence of width.

Advantages:

- learns more features simultaneously.
- Better performance from simple problems.

Disadvantages.

- More memory usage.
- Higher computation cost.
- Can lead to over fitting.